✅ 1. Gumbel distribution

For parameters:

- location: μ
- scale: $\sigma > 0$

$$F(x;\mu,\sigma) = \exp\left(-e^{-(x-\mu)/\sigma}\right), \quad f(x;\mu,\sigma) = \frac{1}{\sigma}e^{-(x-\mu)/\sigma}\exp\left(-e^{-(x-\mu)/\sigma}\right)$$

Define:

$$z = \frac{x-\mu}{\sigma}, \quad e^{-z} = \exp(-(x-\mu)/\sigma)$$

✅ 2. Log-likelihood

Starting from:

$$\mathcal{L} = \prod_{n=1}^{N_T-1} f(x_{l_{n+1}}) \prod_{n=1}^{N_T-1} F(x_{l_n})^{\Delta_{n+1}-1} \times F(x_{l_{N_T}})^{T-l_{N_T}}$$

Taking logs:

$$\ell(\mu,\sigma) = \sum_{n=1}^{N_T-1} \log f(x_{l_{n+1}}) + \sum_{n=1}^{N_T-1} (\Delta_{n+1}-1)\log F(x_{l_n}) + (T-l_{N_T})\log F(x_{l_{N_T}})$$

Plug Gumbel expressions

Log-density:

$$\log f(x) = -\log\sigma - z - e^{-z}$$

Log-CDF:

$$\log F(x) = -e^{-z}$$

Final log-likelihood

Let:

$$z_{n+1} = \frac{x_{l_{n+1}}-\mu}{\sigma}, \quad z_n = \frac{x_{l_n}-\mu}{\sigma}, \quad z_* = \frac{x_{l_{N_T}}-\mu}{\sigma}$$

Then:

$$\boxed{\begin{aligned}\ell(\mu,\sigma) = -(N_T-1)\log\sigma - \sum_{n=1}^{N_T-1} z_{n+1} \\ - \sum_{n=1}^{N_T-1} e^{-z_{n+1}} \\ - \sum_{n=1}^{N_T-1} (\Delta_{n+1}-1)e^{-z_n} \\ - (T-l_{N_T})e^{-z_*}\end{aligned}}$$

✅ 3. First derivatives

We use:

$$\frac{\partial z}{\partial \mu} = -\frac{1}{\sigma}, \quad \frac{\partial z}{\partial \sigma} = -\frac{x-\mu}{\sigma^2}$$

and:

$$\frac{d}{dz}e^{-z} = -e^{-z}$$

✅ Derivative w.r.t. μ

$$\boxed{\begin{aligned}\frac{\partial \ell}{\partial \mu} = \frac{N_T-1}{\sigma} - \frac{1}{\sigma}\sum_{n=1}^{N_T-1} e^{-z_{n+1}} \\ - \frac{1}{\sigma}\sum_{n=1}^{N_T-1} (\Delta_{n+1}-1)e^{-z_n} \\ - \frac{(T-l_{N_T})}{\sigma}e^{-z_*}\end{aligned}}$$

✅ Derivative w.r.t. σ

$$\boxed{\begin{aligned}\frac{\partial \ell}{\partial \sigma} = -\frac{N_T-1}{\sigma} + \frac{1}{\sigma^2}\sum_{n=1}^{N_T-1} (x_{l_{n+1}}-\mu) \\ - \frac{1}{\sigma^2}\sum_{n=1}^{N_T-1} (x_{l_{n+1}}-\mu)e^{-z_{n+1}} \\ - \frac{1}{\sigma^2}\sum_{n=1}^{N_T-1} (\Delta_{n+1}-1)(x_{l_n}-\mu)e^{-z_n} \\ - \frac{(T-l_{N_T})}{\sigma^2}(x_{l_{N_T}}-\mu)e^{-z_*}\end{aligned}}$$

✅ 4. Second derivatives

✅ Second derivative w.r.t. μ

$$\boxed{\begin{aligned}\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{N_T-1}{\sigma^2} - \frac{1}{\sigma^2}\sum_{n=1}^{N_T-1} e^{-z_{n+1}} \\ -\frac{1}{\sigma^2}\sum_{n=1}^{N_T-1}(\Delta_{n+1}-1)e^{-z_n} \\ -\frac{(T-l_{N_T})}{\sigma^2}e^{-z_*}\end{aligned}}$$

✅ Second derivative w.r.t. σ

$$\boxed{\begin{aligned}\frac{\partial^2 \ell}{\partial \sigma^2} = \frac{N_T-1}{\sigma^2} - \frac{2}{\sigma^3}\sum_{n=1}^{N_T-1}(x_{l_{n+1}}-\mu) \\ +\frac{2}{\sigma^3}\sum_{n=1}^{N_T-1}(x_{l_{n+1}}-\mu)e^{-z_{n+1}} \\ +\frac{1}{\sigma^4}\sum_{n=1}^{N_T-1}(x_{l_{n+1}}-\mu)^2 e^{-z_{n+1}} \\ +\frac{2}{\sigma^3}\sum_{n=1}^{N_T-1}(\Delta_{n+1}-1)(x_{l_n}-\mu)e^{-z_n} \\ +\frac{1}{\sigma^4}\sum_{n=1}^{N_T-1}(\Delta_{n+1}-1)(x_{l_n}-\mu)^2 e^{-z_n} \\ +\frac{2(T-l_{N_T})}{\sigma^3}(x_{l_{N_T}}-\mu)e^{-z_*} \\ +\frac{(T-l_{N_T})}{\sigma^4}(x_{l_{N_T}}-\mu)^2 e^{-z_*}\end{aligned}}$$

✅ Cross derivative

$$
\boxed{
\begin{aligned}
\frac{\partial^2 \ell}{\partial \mu \partial \sigma} = &-\frac{N_T - 1}{\sigma^2} \\
&+ \frac{1}{\sigma^2} \sum_{n=1}^{N_T-1} e^{-z_{n+1}} - \frac{1}{\sigma^3} \sum_{n=1}^{N_T-1} (x_{l_{n+1}} - \mu) e^{-z_{n+1}} \\
&+ \frac{1}{\sigma^2} \sum_{n=1}^{N_T-1} (\Delta_{n+1} - 1) e^{-z_n} - \frac{1}{\sigma^3} \sum_{n=1}^{N_T-1} (\Delta_{n+1} - 1)(x_{l_n} - \mu) e^{-z_n} \\
&+ \frac{(T - l_{N_T})}{\sigma^2} e^{-z_*} - \frac{(T - l_{N_T})}{\sigma^3} (x_{l_{N_T}} - \mu) e^{-z_*}
\end{aligned}
}
$$